

CSCE 190: Computing in the Modern World

Class 2 - Careers, Courses

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Carolinian Creed: “I will practice personal and academic integrity.”

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Outline

- Introduction Section
 - Recap from Last Class
- Main Section
 - Understanding an Organization
 - Understanding Jobs and Career Success
 - Research Career
 - Money v/s Success
 - From Course to Careers
 - Course Advising
- Conclusion Section
 - Ask Me Anything

Introduction Section

Recap from Last Lecture - Class 1

We talked about

- Concepts
 - Calculation
 - Computation
 - Decision Support (AI)
- Computation for Running the Class
 - Slack
 - Github
 - Class Management: Google Suite
- Course Details
 - Attendance
 - Grading

Main Section

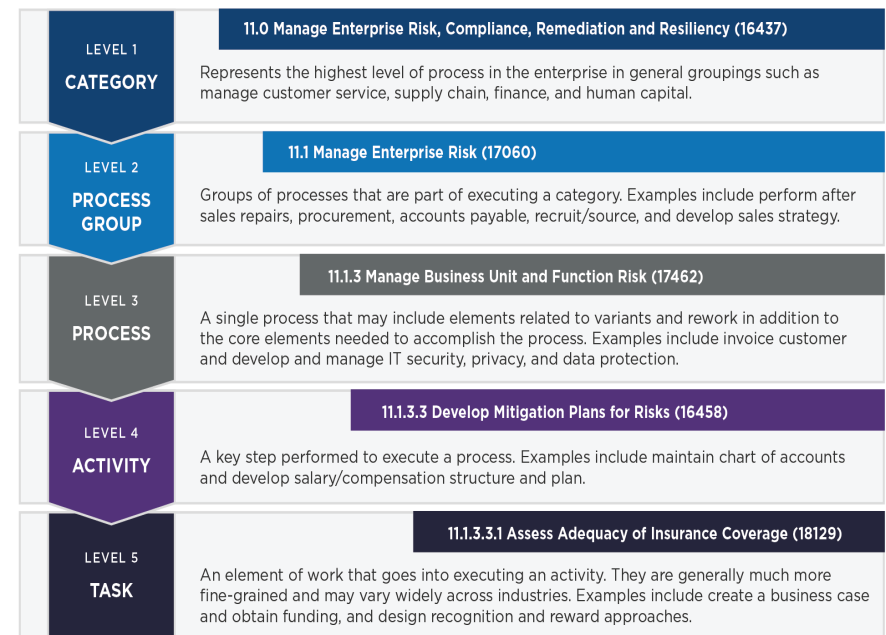
Understanding an Organization

- Any organization, like a company, University or a non-profit, has a set of processes that need to be performed.
- People are hired, and tools are used, to perform these processes.
- Any specific organization, like USC, may have unique processes but usually, most of them are common.
- APQC (American Productivity & Quality Center) helps define organizations, benchmarks them by industries and supports their transformation. Details: <https://www.apqc.org/>
- **Process Classification Framework (PCF)** is a taxonomy from them about business processes that allows organizations to objectively track and compare their performance internally and externally with organizations from any industry.
 - PCF 8.0 in pdf: [https://github.com/biplav-s/course-cmw-f26/blob/main/resources/class2/apqc-K016809-APQC%20Process%20Classification%20Framework%20\(PCF\)%20-%20Cross-Industry%20-%20PDF%20Version%208.0_0.pdf](https://github.com/biplav-s/course-cmw-f26/blob/main/resources/class2/apqc-K016809-APQC%20Process%20Classification%20Framework%20(PCF)%20-%20Cross-Industry%20-%20PDF%20Version%208.0_0.pdf)
 - PCF 8.0 in excel: [https://github.com/biplav-s/course-cmw-f26/blob/main/resources/class2/apqc-K017163-APQC%20Process%20Classification%20Framework%20\(PCF\)%20-%20Utilities%20-%20Excel%20Version%208.0.xlsx](https://github.com/biplav-s/course-cmw-f26/blob/main/resources/class2/apqc-K017163-APQC%20Process%20Classification%20Framework%20(PCF)%20-%20Utilities%20-%20Excel%20Version%208.0.xlsx)

Understanding an Organization: PCF

- Process Classification Framework (PCF).

PCF ID	Hierarchy ID	Name
10002	1.0	Develop Vision and Strategy
10003	2.0	Develop and Manage Products and Services
10004	3.0	Market and Sell Products and Services
20022	4.0	Deliver Physical Products
20025	5.0	Deliver Services
20085	6.0	Manage Customer Service
10007	7.0	Develop and Manage Human Capital
20607	8.0	Manage Information Technology (IT)
17058	9.0	Manage Financial Resources
19207	10.0	Acquire, Construct, and Manage Assets
16437	11.0	Manage Enterprise Risk, Compliance, Remediation, and Resiliency
10012	12.0	Manage External Relationships
10013	13.0	Develop and Manage Business Capabilities



Activity: Explore PCF for Potential Jobs

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- Human Resource => 7.0
- Product Development => 2.0
- Research => 1.0 and 2.0
- Consulting => 5.0
- Corporate Finance => 9.0
- Sales => 3.0, 12.0

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- **External facing**

- Consulting => 5.0
- Sales => 3.0, 12.0
- Deliver Products => 4.0
- Manage Relationship => 12.0

- **Internal facing**

- Human Resource => 7.0
- Product Development => 2.0
- Research => 1.0 and 2.0
- Corporate Finance => 9.0

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Q: Which process is **teaching** at a Univ?

Q: Which process is **research** at a Univ?

What is Career Success?

- **No single answer**
 - Depends on individual and community expectations.
 - Formally studied in psychology, economics, theology, ..., and now even in AI
- But **many indicators of success**
 - Person is happy
 - Internal indicators: money, family, ...
 - External indicators: designation, possessions, ...
 - **Other factors**: personality, luck, ...
 - Person has (financial) independence to do what they want
- Bottomline: **Success is what you define**
 - No one knows all the opportunities and challenges a person faces
 - Comparison with one's peers can be misleading
 - Comparison with one's own expectations and performance is more actionable
 - In AI, success concept is studied with the notion of rational decision making with limited resources

Success v/s Money

- Money: possible due to a number of factors
 - Related: consequence of success
 - Unrelated: e.g., winning a lottery
- Planned Success
 - Based on an individual's knowledge and actions
- Career Success v/s Life Success
 - Many people find life success outside career

Course to Careers – A Perspective

- Jobs are continuously evolving
 - Many jobs your grand parents had were **not** available to your parents, but they had others
 - Likewise, many opportunities your parents had will **not** be available to you, but you will have others
- You can see job descriptions at US Bureau of Labor Statistics [Occupational Outlook Handbook](#) and the [Standard Occupational Classification \(SOC\)](#) system
 - OOH - <https://www.bls.gov/ooh/>
 - SOC - <https://www.bls.gov/soc/>

Course to Careers – A Perspective

OOH - <https://www.bls.gov/ooh/>

[OOH HOME](#) | [OCCUPATION FINDER](#) | [OOH FAQ](#) | [HOW TO FIND A JOB](#) | [A-Z INDEX](#) | [OOH SITE MAP](#)

Search Handbook [Go](#)

OCCUPATIONAL OUTLOOK HANDBOOK

Environmental Engineers

[Summary](#) | [What They Do](#) | [Work Environment](#) | [How to Become One](#) | [Pay](#) | [Job Outlook](#) | [State & Area Data](#) | [Similar Occupations](#) | [More Info](#)

Summary

Quick Facts: Environmental Engineers	
2024 Median Pay ?	\$104,170 per year \$50.08 per hour
Typical Entry-Level Education ?	Bachelor's degree
Work Experience in a Related Occupation ?	None
On-the-job Training ?	None
Number of Jobs, 2024 ?	39,400
Job Outlook, 2024–34 ?	4% (As fast as average)
Employment Change, 2024–34 ?	1,500



What Environmental Engineers Do

Environmental engineers use engineering disciplines in developing solutions to problems of planetary health.

Work Environment

Environmental engineers work both in an office setting and in the field. Most work full time, and some work more than 40 hours per week.

How to Become an Environmental Engineer

Environmental engineers typically need a bachelor's degree in environmental engineering or a related field, such as civil, chemical, or general engineering. Some employers prefer to hire candidates who have gained practical experience in an internship or a cooperative education program.

Course to Careers – A Perspective

SOC - <https://www.bls.gov/soc/>

SOC (2018) - https://github.com/biplav-s/course-cmw-f26/blob/main/resources/class2/soc_structure_2018.pdf

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www.bls.gov/soc/2018/soc_structure_2018.pdf

Automatic Zoom

Major Group	Minor Group	Broad Group	Detailed Occupation
19-0000			Life, Physical, and Social Science Occupations
	19-1000		Life Scientists
		19-1010	Agricultural and Food Scientists
		19-1011	Animal Scientists
		19-1012	Food Scientists and Technologists
		19-1013	Soil and Plant Scientists
		19-1020	Biological Scientists
		19-1021	Biochemists and Biophysicists
		19-1022	Microbiologists
		19-1023	Zoologists and Wildlife Biologists
		19-1029	Biological Scientists, All Other
		19-1030	Conservation Scientists and Foresters
		19-1031	Conservation Scientists
		19-1032	Foresters
		19-1040	Medical Scientists
		19-1041	Epidemiologists
		19-1042	Medical Scientists, Except Epidemiologists
		19-1090	Miscellaneous Life Scientists
		19-1099	Life Scientists, All Other
	19-2000		Physical Scientists
		19-2010	Astronomers and Physicists
		19-2011	Astronomers
		19-2012	Physicists
		19-2020	Atmospheric and Space Scientists
		19-2021	Atmospheric and Space Scientists
		19-2030	Chemists and Materials Scientists
		19-2031	Chemists
		19-2032	Materials Scientists
		19-2040	Environmental Scientists and Geoscientists
		19-2041	Environmental Scientists and Specialists, Including Health
		19-2042	Geoscientists, Except Hydrologists and Geographers
		19-2043	Hydrologists
		19-2090	Miscellaneous Physical Scientists
		19-2099	Physical Scientists, All Other

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Information Technology (IT) Careers

- Management
- Research
- Development / Production
- Sales
- Services/ Consulting
- Support

Development activities

- Software development:
- Hardware development:
- Services delivery
- Solutions creation: consists of software, hardware, services

IT Research Career

- Need to deal with unknowns
- Need to deal with no-s, can't be done
- Read a lot, write a lot
- Can make your point via
 - Doing experiments
 - Proving via theory
 - Connecting with related work
- High risk, high reward
- Good for those who have **self-confidence**

Research can be in any IT area

- Software development:
- Hardware development:
- Services delivery
- Solutions creation: consists of software, hardware, services

Courses

- Is a university's way of imparting skills in well structured modules
- Completing courses leads to meeting graduation requirements for a degree
- **Advising** is the structured way to get advice

Advisement

- Pre-advisement checklist:
 - https://sc.edu/about/offices_and_divisions/advising/student_advising_resources/pre_advisement/index.php
- Advisement FAQ:
 - <https://www.cse.sc.edu/undergraduate/advisement>
- Major maps (all required courses):
 - https://sc.edu/about/offices_and_divisions/advising/advisor_toolbox/majormaps.php?search=2026-2027
 - Look for the major you are interested in
 - For Computer Science, see a local copy at: https://github.com/biplav-s/course-cmw-f26/blob/main/resources/class2/2026_cs_map.pdf

Activity – Writing Mission Statement

- Objective: identifying your **personal** mission for a career and success
- Guideline:
 - Think of short (1-3 years), medium (4-7 years) and long-term (8 -) goals
 - While career is the focus, feel free to include personal goals as well
 - Define success criteria for all goals
 - Try to structure how short term (e.g., undergrad program) may help meet **medium-term goals**
 - Try to structure how short and medium may help meet **long-term goals**
- Try to identify organizations and job roles of interest
- Strategize how your major and courses may help

Concluding Section

Conclusion

- We tried to understand organizations, business processes and job roles
- Discussed IT and Research Careers
- We looked at success and career success indicators
- We talked about courses and advising

Week	Content	Assignment
1	Intro, syllabus, university resources	Slack account
2	CSE majors, career paths, research areas	Write a mission statement
3	Github (account and basics)	Github account, repo
4	VS Code, HTML basics	None
5	More HTML, CSS	Update Github with CSS
6	Brainstorming, problem statement	Problem statement
7	Computing – (Software) Dev. Life Cycle	Draft Requirement, Specification
8	Computing – Testing a System	Test Plan
9	Building a resume, LinkedIn	Draft resume in PDF
10	Review resumes	Review resumes, update Github
11	Convert resume to HTML	Begin conversion, update Github
12	Github code repos	Add repos to portfolio, add SDLC outcomes
13	CSE research areas	(Finalize website)
N/A	Last day of classes: 4 Dec, 2026	All extensions end

Ask Me Anything
